Exercise Sheet 1

Uncertainty, Decisions and Reasoning

Probabilistic Machine Learning

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1.A Decisions From Probabilities

Exercise 1.A.1: Three Loan Applicants

A loan model gives three applicants a **default probability** of 0.05, 0.35 and 0.45. Accepting an applicant who defaults costs the bank **€8,000**; rejecting one who would have repaid costs **€500** in lost margin.

1. Which applicants does the rule "reject if $p > 0.5$" reject?
2. Compute the expected cost of *accepting* and of *rejecting* each applicant. Which decision minimises it?
3. Derive the break-even probability p^* in general, for a cost c_A of accepting a defaulter and c_R of rejecting a good customer. Evaluate it here.
4. How does p^* change if the bank's margin doubles? If the loan is ten times larger (and the margin stays)? Explain the direction in one sentence each.

Exercise 1.A.1: Three Loan Applicants — Solution

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1. **Nobody** — all three are below 0.5.
2. Accept: $p \cdot 8{,}000$; reject: $(1-p) \cdot 500$.

p	accept	reject	decision
0.05	€400	€475	accept (just)
0.35	€2,800	€325	reject
0.45	€3,600	€275	reject

3. Reject when $p\,c_A > (1-p)\,c_R$, i.e. $p > p^* = \dfrac{c_R}{c_R + c_A} = \dfrac{500}{8{,}500} \approx 0.059$.
4. Margin doubled: $p^* = 1{,}000/9{,}000 \approx 0.111$ — a lost customer costs more, so the bank accepts more risk. Loan $\times 10$: $p^* = 500/80{,}500 \approx 0.006$ — a default costs more, so the bank rejects almost everyone with any risk. The model output never changed.

Exercise 1.A.2: Allow, Block — or Ask a Human?

A fraud model scores a transaction with probability p of fraud. Three actions:

- **allow**: if it is fraud, the bank loses **€900**;
- **block**: if it is legitimate, an annoyed customer costs **€300**;
- **review**: an analyst checks it for a flat **€60** and always decides correctly.

1. Write the expected cost of each action as a function of p.
2. For which range of p is each action optimal?
3. Without a review option, where would the threshold between allow and block be? What does the review option change?

Exercise 1.A.2: Allow, Block — or Ask a Human? — Solution

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1. Allow: $900\,p$. Block: $300\,(1-p)$. Review: 60.
2. Allow beats review when $900\,p < 60$, i.e. $p < 1/15 \approx 0.067$; block beats review when $300\,(1-p) < 60$, i.e. $p > 0.8$. So: **allow** for $p < 0.067$, **review** for $0.067 < p < 0.8$, **block** for $p > 0.8$.
3. Without review: allow $=$ block at $900\,p = 300\,(1-p), p = 0.25$. With review, the whole uncertain middle goes to a human — "defer deliberately" from the lecture, with its range computed rather than guessed. A label without a probability could not support this three-way policy at all.

Exercise 1.A.3: Choosing a Threshold From Data (Python)

```python
import numpy as np
rng = np.random.default_rng(1)
p = rng.beta(0.5, 6, size=10_000)        # model scores (assume calibrated)
y = rng.random(10_000) < p               # true labels drawn from the scores
c_A, c_R = 8_000, 500                    # costs as in 1.A.1
```

1. For thresholds $t \in \{0, 0.01, \ldots, 1\}$ ("reject if $p > t$"), compute the **average realised cost** per applicant.
2. Which t minimises it? Compare with p^* from 1.A.1.
3. What would happen to the best t if the scores were systematically too low (say `p_reported = p / 2`)?

Exercise 1.A.3: Choosing a Threshold From Data — Solution

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```
ts = np.linspace(0, 1, 101)
cost = [np.mean(np.where(p > t, (~y) * c_R, y * c_A)) for t in ts]
print(ts[np.argmin(cost)])               # ≈ 0.05
```

2. The minimum lies at $t \approx 0.05$–0.06 (with this seed 0.05, and 0.06 within a fraction of a euro), next to the break-even $p^* \approx 0.059$: with calibrated scores, the cost-optimal threshold is the one computed from the costs alone.
3. With `p_reported = p / 2` the best threshold on the *reported* scores halves to about 0.03 — the formula p^* is only right for calibrated scores. This is why Lecture 04 asks when a score deserves to be read as a probability.

1.B Where Uncertainty Comes From

Exercise 1.B.1: Noise or Ignorance?

For each case, say whether the uncertainty is mainly **aleatoric** (in the problem), **epistemic** (in our knowledge), or both — and whether more data of the same kind would help.

1. Predicting the outcome of a fair coin flip.
2. A house-price model trained on flats of 40–120 m^2, asked about a 600 m^2 villa.
3. A spam filter trained on English mail, applied to Finnish mail.
4. Two customers with identical records; one churns, one does not.
5. A clinical measurement with known device noise, taken once per patient.

Exercise 1.B.1: Noise or Ignorance? — Solution

Exercise 1.B.1: Noise or Ignorance? — Solution

1. **Aleatoric.** No amount of data predicts a fair flip.
2. **Epistemic.** No data near 600 m^2; villas in the training data would help.
3. **Epistemic** — and a shift: the model has not seen the language. Finnish training mail helps; more English mail does not.
4. **Aleatoric relative to the recorded attributes.** More rows with the same attributes do not help; *more attributes* might — "inherent" is always relative to what is measured.
5. **Aleatoric** for a single reading. Repeating the measurement reduces the uncertainty about the patient's true value (the average's standard error falls), not the device noise itself.

Exercise 1.B.2: One Pipeline, Many Uncertainties

A bank trains a logistic-regression fraud model on last year's transactions. Labels come from chargebacks, which arrive up to 90 days after the transaction. The test set holds 2,000 recent transactions, of which **3** are fraud.

Name **five** distinct places where uncertainty enters this system. For each, say which box of the learning system it lives in and propose one mitigation.

Exercise 1.B.2: One Pipeline, Many Uncertainties — Solution

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source	box	mitigation
fraud looks legitimate (aleatoric)	world	route the grey zone to review (1.A.2)
labels late and incomplete	data	label only after 90 days
last year ≠ this year (drift)	data	monitor scores; retrain
linear model misses interactions	model	compare with a richer model
few frauds, loose parameters (epistemic)	parameters	more positives; regularise
3 test frauds: recall barely measurable	evaluation	more test frauds; report an interval

Any five count. The last is the one most often missed: with 3 frauds, a measured recall can only be 0, ⅓, ⅔ or 1.

1.C Reasoning Under Uncertainty

Exercise 1.C.1: Which Mode of Reasoning?

Classify each step as **deduction**, **induction** or **abduction**.

1. "All training examples with a foreign IP and a night-time purchase were fraud, so such transactions are fraud."
2. "The model's accuracy dropped last week; the most plausible explanation is the new payment provider."
3. "If the logistic model is correct, doubling feature x_1 multiplies the odds by $e^{w_1 x_1}$."
4. "Customers who use the app churn less; perhaps loyal customers install the app."
5. "Gradient descent with this learning rate on this convex loss converges."

Exercise 1.C.1: Which Mode of Reasoning? — Solution

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1. **Induction** — a regularity generalised from observed cases (and an over-confident one).
2. **Abduction** — the best available explanation of an observation.
3. **Deduction** — follows from the model's definition; true *if* the model is.
4. **Abduction** — one explanation of a correlation, with a rival (the app causes loyalty) that the data alone cannot rule out.
5. **Deduction** — a theorem about the algorithm, given its premises.

Exercise 1.C.2: Save the Model or Change It?

A fraud model has been reliable for two years. After a new product launch, its recall on the new product's transactions falls from 0.8 to 0.4.

1. Name the "save the theory" and the "change the theory" move in this setting.
2. Design two cheap checks, in order, that would tell the moves apart.
3. What evidence would make you abandon the current model family — and why should one bad week not be enough?

Exercise 1.C.2: Save the Model or Change It? — Solution

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1. **Save:** the model is right but has not seen the new product — collect and label its transactions, retrain (epistemic). **Change:** fraud on the new product follows patterns the features or the model family cannot express.
2. First check that nothing trivial broke (a feature missing or mis-encoded for the new product). Then add labelled examples and watch the learning curve on the new product: if recall recovers, it was data; if it stays flat, it is the model or noise (compare with a richer model or new features).
3. A flat learning curve *and* a rival — new features or a model class — that explains the new cases better on held-out data. One bad week is a small sample: its recall has a wide interval, and after two good years the prior on "the model is fine" is strong — Uranus before Mercury.